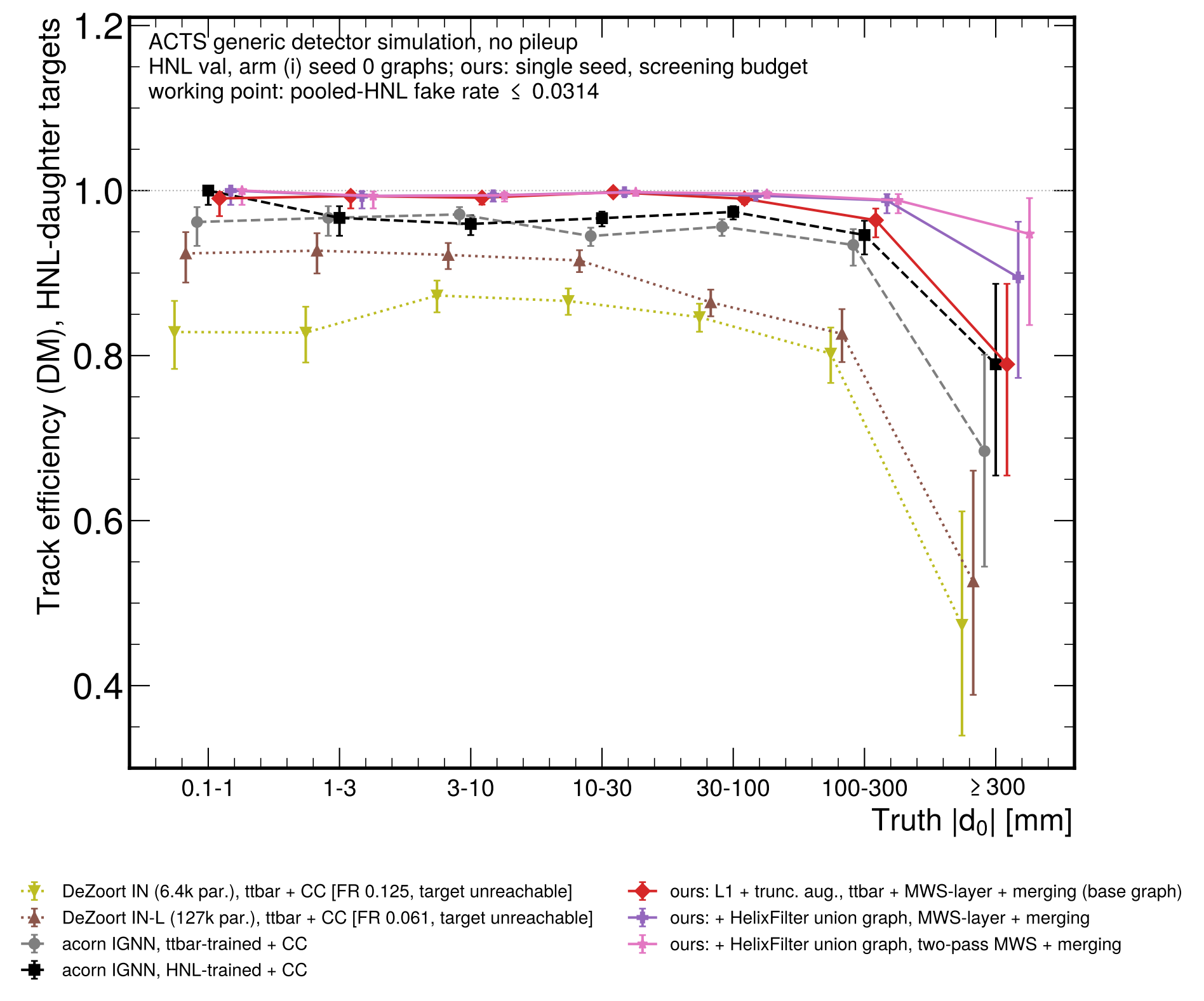
ACTS generic detector simulation, no pileup
HNL val, arm (i) seed 0 graphs; ours: single seed, screening budget
working point: pooled-HNL fake rate ≤ 0.0314
Track efficiency (DM), HNL-daughter targets
1.2
1.0
0.8
0.6
0.4
0.1-1
1-3
3-10
10-30
30-100
100-300
≥ 300
Truth |d0| [mm]
DeZoort IN (6.4k par.), ttbar + CC [FR 0.125, target unreachable]
DeZoort IN-L (127k par.), ttbar + CC [FR 0.061, target unreachable]
acorn IGNN, ttbar-trained + CC
acorn IGNN, HNL-trained + CC
ours: L1 + trunc. aug., ttbar + MWS-layer + merging (base graph)
ours: + HelixFilter union graph, MWS-layer + merging
ours: + HelixFilter union graph, two-pass MWS + merging